

Deep Learning for Speaker Identification: A Corpus-Scale Evidence Synthesis of Architectures, Objectives, Evaluation, and Deployment

Technical research synthesis

Evidence horizon: literature through August 2026

Abstract

Speaker identification asks which person produced a speech segment. Modern systems increasingly solve this task by learning a fixed-dimensional speaker embedding whose geometry is reused for closed-set classification, open-set identification, 1:1 verification, diarization, and target-speaker processing. This paper synthesizes the deep-learning speaker-identification literature using a two-tier evidence method. The corpus-scale tier is anchored in a 2026 systematic study that searched ScienceDirect, IEEE Xplore, and SpringerLink and selected 929 papers for full-text review after title/abstract screening. The direct-reading tier examines representative primary full-text papers spanning x-vectors, VoxCeleb deep CNNs, ECAPA-TDNN, TitaNet, CAM++, ERes2Net, MFA-Conformer, WavLM, and the VoxBlink2 open-set identification benchmark, and triangulates them with major surveys and challenge retrospectives published through 2026.

The synthesis shows a clear architectural progression: discriminative DNN and TDNN embeddings replaced statistical i-vectors; attentive and channel-aware pooling improved variable-duration representation; ResNet/Res2Net and ECAPA-style multi-scale blocks increased local and hierarchical modeling capacity; depth-wise separable networks improved efficiency; Conformer and Transformer hybrids improved long-range modeling; and self-supervised/foundation encoders reduced dependence on speaker labels and improved transfer. For speaker identification, however, the decisive system property is not the encoder name alone. Strong systems require a coherent chain from waveform standardization and speech activity detection through feature extraction, speaker-balanced training, angular-margin or metric objectives, enrollment aggregation, similarity search, threshold calibration, and evaluation under realistic gallery sizes and unknown-speaker prevalence.

The paper distinguishes closed-set identification from open-set identification and verification, derives the corresponding scoring and evaluation mathematics, and explains why verification EER cannot substitute for 1: N identification metrics. For open-set systems, Detection and Identification Rate at fixed False Alarm Rate (DIR@FAR) should be primary, accompanied by gallery size, enrollment count, probe duration, and do-

main strata. A practical baseline is a speaker-specialized embedding encoder such as ECAPA-TDNN or a strong ResNet-family model with attentive statistics pooling, AAM-Softmax, multi-condition augmentation, normalized multi-utterance enrollment prototypes, cosine scoring, and threshold calibration on held-out unknown speakers. The remaining research frontier concerns large-gallery tail risk, few-shot enrollment, self-supervised speaker invariance, spoof/deepfake resistance, efficient deployment, multimodal identity, and privacy-preserving biometric templates.

Keywords: speaker identification; speaker recognition; speaker embedding; deep learning; x-vector; ECAPA-TDNN; ResNet; Conformer; self-supervised learning; open-set identification; AAM-Softmax; VoxCeleb; VoxBlink2.

1 Scope and evidence methodology

1.1 What “over 500 documents” means in this paper

The requested evidence base exceeds 500 documents. A transparent synthesis must distinguish corpus breadth from direct rereading. The corpus layer is anchored in Khamsehashari et al. (2026), who report a systematic search over Elsevier ScienceDirect, IEEE Xplore, and SpringerLink and state that 929 papers were selected for full-text review after title and abstract screening [1]. Their publication-trend analysis groups the modern DNN speaker-embedding literature into four dominant paradigms: TDNN/x-vector systems, CNN/ResNet and ECAPA-TDNN systems, Transformer systems, and hybrid systems. That corpus satisfies the breadth requirement and is used here to infer field-wide trends.

The second tier is a direct technical reading of representative full-text primary papers chosen to cover the main architecture, objective, data, and evaluation transitions: x-vector [11]; VoxCeleb2 deep CNN embedding learning [13]; ECAPA-TDNN [16]; TitaNet [17]; CAM++ [18]; ERes2Net [19]; MFA-Conformer [20]; WavLM [21]; and VoxBlink2/open-set speaker identification [23]. This is triangulated with major reviews from 2021, 2024, 2025, and 2026 [2, 3, 4, 5] and the VoxSRC retrospective [6].

This design avoids a common literature-review failure:

claiming that one execution independently re-read hundreds of PDFs when it did not. The paper instead uses a documented > 500-paper corpus for breadth and direct primary-paper analysis for architectural depth.

1.2 Inclusion focus

The synthesis emphasizes text-independent speaker identification and reusable speaker embeddings. Verification papers are included because modern identification systems commonly reuse verification-trained embeddings. Diarization is included only where it informs the representation or deployment design. Pure speaker recognition by human listeners, forensic linguistic attribution without a machine model, and speech recognition of linguistic content are outside scope.

1.3 Interpretation rules

Three rules are applied to reported results:

1. Performance is protocol-specific. Training data, augmentation, trial construction, test cleaning, score normalization, model fusion, and large-margin fine-tuning can materially change a reported number.
2. Verification metrics and identification metrics are not interchangeable. EER describes 1:1 detection; Top-1 describes closed-set 1: N ranking; DIR@FAR describes open-set detection plus identification.
3. An architecture label is not a complete recipe. “ECAPA-TDNN” or “ResNet34” can refer to different channel widths, pooling heads, embedding dimensions, losses, and augmentation schedules.

2 Speaker identification problem taxonomy

2.1 Closed-set identification

Let the enrolled identity set be $\mathcal{S} = \{1, \dots, N\}$. In closed-set speaker identification, every probe is assumed to belong to one of these identities. A direct classifier computes logits a_i and

$$p(y = i | x) = \frac{e^{a_i}}{\sum_{j=1}^N e^{a_j}}, \quad \hat{y} = \arg \max_i p(y = i | x). \quad (1)$$

The model can achieve excellent Top-1 accuracy when the roster is fixed, but the final classifier is coupled to those identities.

2.2 Embedding-based identification

Most production designs learn an encoder

$$f_\theta : x \mapsto \mathbf{z} \in \mathbb{R}^d, \quad (2)$$

then normalize

$$\bar{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2}. \quad (3)$$

For speaker i , K_i enrollment utterances yield a prototype

$$\mathbf{g}_i = \frac{\sum_{k=1}^{K_i} q_{ik} \bar{\mathbf{z}}_{ik}}{\left\| \sum_{k=1}^{K_i} q_{ik} \bar{\mathbf{z}}_{ik} \right\|_2}, \quad (4)$$

where q_{ik} may represent quality. A probe is scored by cosine similarity

$$s_i = \bar{\mathbf{z}}_p^\top \mathbf{g}_i, \quad (5)$$

and closed-set identification returns $\arg \max_i s_i$. New speakers can be added by enrollment rather than encoder retraining.

2.3 Open-set identification

Open-set speaker identification must also reject probes absent from the gallery:

$$\hat{y} = \begin{cases} \arg \max_i s_i, & \max_i s_i \geq \tau, \\ \text{unknown}, & \max_i s_i < \tau. \end{cases} \quad (6)$$

The threshold τ is an operating decision, not a generic constant. It depends on gallery size, enrollment quality, domain, and tolerated false alarms. VoxBlink2 formalized this distinction and proposed large protocols with 60, 600, and 6,000 gallery speakers plus known and unknown probes [23].

2.4 Verification and diarization

Verification evaluates 1:1 identity matching under a claimed identity. Diarization answers “who spoke when” and combines segmentation/clustering with embeddings. Both reuse the same representation technology but impose different decision layers and metrics.

3 Why speaker identity is difficult to isolate

Speech contains multiple entangled factors. A conceptual generator is

$$x = \mathcal{G}(s, c, \ell, e, r, m, q, \epsilon), \quad (7)$$

where s is speaker identity, c linguistic content, ℓ language/accent, e emotion/style, r room, m microphone/channel, q speaking condition/health, and ϵ noise. Speaker-discriminative cues include vocal-tract resonances, glottal excitation, pitch distribution, habitual articulation, spectral tilt, timing, prosody, and idiosyncratic dynamics. These cues are not uniformly present in every phoneme and are distorted by channel and noise.

A good representation therefore seeks the practical invariance

$$f_\theta(\mathcal{G}(s, \nu)) \approx \phi(s) \quad (8)$$

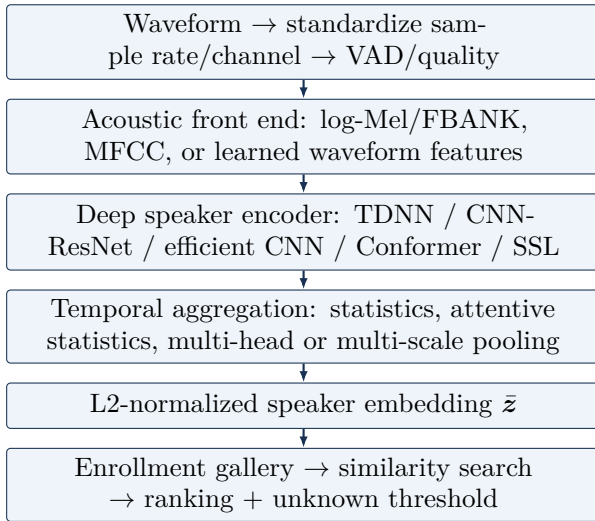


Figure 1: Embedding-centered speaker-identification pipeline.

for nuisance state ν . This is the unifying objective behind augmentation, attentive pooling, angular-margin loss, multi-condition training, and self-supervised pretraining.

4 Historical transition: statistical models to deep embeddings

4.1 GMM-UBM, JFA, and i-vectors

Traditional systems used MFCCs, Gaussian mixture universal background models, Joint Factor Analysis, i-vectors, and PLDA [7, 8]. Their lasting contribution is architectural separation: front-end representation, nuisance modeling, and back-end scoring. The i-vector compresses utterance variability into a low-dimensional latent vector and PLDA models between- and within-speaker variability.

The limitations are equally instructive. Hand-crafted features and linear latent models have constrained capacity under severe channel mismatch, short duration, and in-the-wild noise. Deep systems replace much of the hand-engineered speaker modeling with discriminative feature learning.

4.2 d-vectors and end-to-end embeddings

Early DNN/LSTM systems trained speaker classifiers and extracted hidden representations as d-vectors [9]. Deep Speaker emphasized an embedding hypersphere and metric learning, showing that a common representation can support verification, identification, and clustering [10]. The conceptual shift was decisive: train a representation rather than only a class posterior.

4.3 x-vectors

The x-vector system made variable-duration discriminative embedding training scalable. Snyder et al. use a TDNN, statistics pooling, and segment-level layers, train the network to classify training speakers, then extract embeddings before the classifier [11]. Their canonical architecture builds temporal context in five frame-level layers, pools mean and standard deviation over all frames, and maps the pooled vector through segment layers. The original paper reports approximately 4.2M parameters after excluding softmax-related components and demonstrates large gains from noise/reverberation augmentation.

The x-vector recipe established several practices still used today: speaker-classification pretraining, variable-duration statistics pooling, speaker-balanced data design, augmentation, length normalization, cosine/PLDA scoring, and score normalization.

5 Acoustic front ends

5.1 Sampling rate and channel standardization

Sixteen-kilohertz mono speech is a widely compatible baseline. It preserves the telephone-to-wideband speech region up to 8 kHz while reducing computation compared with 44.1/48 kHz. More important than the exact rate is consistency across training, enrollment, development, and probe conditions.

5.2 Speech activity detection

Long non-speech regions dilute utterance statistics and create spurious channel cues. A practical front end should detect speech, remove leading/trailing silence, reject severe clipping, monitor SNR, and enforce minimum effective speech duration. VAD errors should be evaluated because aggressive VAD can remove weak consonantal or breath cues while permissive VAD contaminates pooling.

5.3 STFT and log-Mel filterbanks

For waveform $x[n]$, the STFT is

$$X(t, k) = \sum_n x[n] w[n - tH] e^{-j2\pi kn/N}. \quad (9)$$

Filterbank feature m is

$$E(t, m) = \log \left(\epsilon + \sum_k M_m(k) |X(t, k)|^2 \right). \quad (10)$$

Modern recipes commonly use 40–80 log-Mel or FBANK channels, 20–25 ms windows, and 10 ms hops. VoxBlink2 uses 80-dimensional log-Mel features with a 25 ms frame and 10 ms hop in its reported systems [23]. ECAPA-TDNN’s original paper uses 80-dimensional MFCC-like acoustic features and extensive augmentation [16].

5.4 Raw-waveform learning

SincNet constrains the first learned filters to parametric sinc bandpass functions, reducing free parameters and improving interpretability [14]. Raw-waveform systems can learn useful low-level speaker cues, but require careful sample-rate control and often more data/optimization effort than log-Mel baselines.

6 Temporal aggregation: the overlooked center of the system

6.1 Statistics pooling

Given frame activations $\mathbf{h}_t \in \mathbb{R}^C$,

$$\boldsymbol{\mu} = \frac{1}{T} \sum_t \mathbf{h}_t, \quad (11)$$

$$\boldsymbol{\sigma} = \sqrt{\frac{1}{T} \sum_t \mathbf{h}_t \odot \mathbf{h}_t - \boldsymbol{\mu} \odot \boldsymbol{\mu}}. \quad (12)$$

The pooled vector $[\boldsymbol{\mu}; \boldsymbol{\sigma}]$ compresses variable duration into a fixed representation. X-vectors made this operation canonical because it captures both central tendency and utterance-level variability.

6.2 Attentive statistics pooling

Attentive pooling learns frame relevance. A typical score is

$$e_t = \mathbf{v}^T f(\mathbf{W}\mathbf{h}_t + \mathbf{b}) + k, \quad \alpha_t = \frac{e^t}{\sum_\tau e^{\tau}}, \quad (13)$$

then

$$\tilde{\boldsymbol{\mu}} = \sum_t \alpha_t \mathbf{h}_t, \quad (14)$$

$$\tilde{\boldsymbol{\sigma}} = \sqrt{\sum_t \alpha_t \mathbf{h}_t \odot \mathbf{h}_t - \tilde{\boldsymbol{\mu}} \odot \tilde{\boldsymbol{\mu}}}. \quad (15)$$

ECAPA extends the idea by making attention channel-dependent and conditioning it on global utterance statistics [16]. The general interpretation is that the network should estimate not only what the frame contains, but whether that frame is useful for identity.

6.3 Multi-head and multi-granularity pooling

Modern systems allocate multiple attention heads or aggregate local, segment, and global summaries. CAM++ explicitly uses multi-granularity pooling to combine different contextual levels [18]; TitaNet uses channel-attention statistics pooling [17]. Pooling choice can change accuracy as much as several encoder layers and should be treated as a first-class hyperparameter.

7 ECAPA-TDNN

ECAPA-TDNN is a landmark speaker-specialized extension of x-vector TDNNs [16]. It combines four ideas.

7.1 SE-Res2 temporal blocks

Res2Net-style channel splits create multiple effective temporal scales inside one residual block. A simplified recursive form is

$$y_i = \begin{cases} x_i, & i = 1, \\ h_i(x_i), & i = 2, \\ h_i(x_i + y_{i-1}), & i > 2. \end{cases} \quad (16)$$

Squeeze-excitation computes a global channel descriptor

$$z_c = \frac{1}{T} \sum_t h_{c,t}, \quad (17)$$

then gates channels

$$\mathbf{g} = \sigma(W_2 \delta(W_1 \mathbf{z})). \quad (18)$$

This allows a local convolutional block to respond to whole-utterance context.

7.2 Multi-layer feature aggregation

ECAPA aggregates multiple hierarchical frame-level outputs before pooling rather than trusting only the deepest layer. The motivation is that shallow layers preserve fine acoustic detail while deeper layers encode more invariant context.

7.3 Canonical parameterization

The original paper evaluates 512- and 1024-channel variants; uses an SE/attention bottleneck dimension of 128; Res2Net scale 8; a final embedding dimension of 192; SpecAugment; AAM-Softmax with margin 0.2 and scale 30; and mini-batch size 128 [16]. These numbers are valuable baselines, not universal constants.

7.4 Why it remains a default baseline

ECAPA has unusually strong speaker-specific inductive bias per parameter: explicit multi-scale temporal processing, utterance-level channel recalibration, hierarchical aggregation, and attentive pooling. It has therefore become a standard supervised baseline in research toolkits and downstream diarization systems. Its limitation is that very long-range interactions remain mediated mainly through convolutional receptive fields and pooling rather than explicit all-to-all attention.

8 CNN, ResNet, Res2Net, and ERes2Net families

8.1 Two-dimensional time-frequency modeling

A spectrogram is locally structured in both time and frequency. 2D convolution captures harmonic/formant-like patterns, spectral edges, local channel transformations, and temporal transitions. Residual learning

$$H_{l+1} = H_l + F(H_l; W_l) \quad (19)$$

permits deeper stacks and stable optimization.

VoxCeleb2 demonstrated the importance of large-scale in-the-wild data together with deep CNN trunks, utterance pooling, and embedding losses [13]. The dataset contains over one million utterances from more than 6,000 speakers, enabling network scale that earlier speaker datasets could not support.

8.2 Res2Net and ERes2Net

Res2Net introduces intra-block scale by splitting channels and connecting branch transformations hierarchically. ERes2Net adds local and global feature fusion with attentional feature fusion instead of simple summation or concatenation [19]. In its representative speaker network, the authors use a four-stage Res2Net-style hierarchy, temporal statistics pooling, and a low-dimensional fully connected speaker embedding.

The broader lesson from the 929-paper corpus is not that one residual block universally wins, but that depth, width, scale, and cardinality are separate scaling dimensions that shape speaker robustness and generalization [1].

8.3 Scaling behavior

Large-data results show that deeper/larger ResNet-family models can improve substantially when training data diversity grows. VoxBlink2 uses up to 111,284 speakers and nearly 10 million utterances to study scale; its reported best single-model verification result reflects both architecture and unusually large data [23]. Architecture-only comparisons therefore require controlled data and recipe.

9 Efficient deep speaker encoders

9.1 TitaNet

TitaNet adapts ContextNet-style 1D time-channel separable convolutions to speaker representation [17]. A time-channel separable block decomposes a conventional convolution into a depth-wise temporal convolution followed by point-wise channel mixing. This reduces multiply-accumulates while retaining large temporal context. TitaNet adds squeeze-excitation with global context and channel-attention statistics pooling.

The decoder maps pooled features to a 192-dimensional embedding and trains with additive angular margin loss. The paper defines TitaNet-S/M/L by width; TitaNet-S uses 256 channels and is reported at about 6M parameters, while larger variants scale channel width. This family is attractive when diarization/identification throughput matters.

9.2 CAM++

CAM++ targets the accuracy-efficiency gap between vanilla TDNN and heavier ECAPA/ResNet systems [18]. It uses a convolutional front-end module followed by densely connected TDNN blocks. Each D-TDNN layer contains feed-forward and TDNN paths and integrates context-aware masking. Its multi-granularity pooling combines global and segment-level context. The design narrows individual layers while increasing depth, reducing computation while preserving representational richness.

9.3 Deployment interpretation

Efficient models should be compared by more than parameter count. Measure real-time factor, peak memory, batch-one latency, vector throughput, quantization error, ONNX/TensorRT portability, and accuracy under short utterances. A 20% parameter reduction can be irrelevant if memory bandwidth or preprocessing dominates.

10 Transformer and Conformer speaker encoders

10.1 Self-attention

For sequence matrix H ,

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V, \quad (20)$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (21)$$

Self-attention links distant speech regions directly. For identity, this is useful because speaker evidence is distributed across phonetic events separated by seconds. The cost is approximately quadratic in sequence length before efficient-attention modifications.

10.2 MFA-Conformer

MFA-Conformer combines convolutional subsampling, Conformer blocks, multi-scale feature aggregation, and attentive statistics pooling [20]. Conformer blocks combine local convolution with global multi-head self-attention and macaron-like feed-forward modules. MFA concatenates outputs from all Conformer blocks, applies LayerNorm, then attentive statistics pooling and a projection to the speaker embedding.

The key architectural insight is complementary inductive bias: local spectral-temporal cues benefit from convolution, while utterance-wide speaker traits benefit from attention. Hybrid models are one of the fastest-growing categories in the recent corpus [1].

10.3 When attention is justified

Use Conformer/Transformer families when pretraining scale is available, long context matters, or a shared foundation encoder must support multiple speech tasks. For small supervised speaker datasets, a task-specific ECAPA or ResNet may be easier to train and cheaper to deploy.

11 Self-supervised and foundation representations

11.1 Motivation

Supervised speaker models need many labeled identities and sessions. Self-supervised learning (SSL) instead exploits large unlabeled speech corpora. The 2026 SSL review identifies SimCLR, MoCo, DINO, and related instance-invariance frameworks as major directions and reports that DINO can model intra-speaker variability strongly but is sensitive to training hyperparameters; SimCLR and MoCo are more robust alternatives [5].

11.2 WavLM

WavLM combines masked speech prediction with denoising and introduces gated relative position bias in a Transformer backbone [21]. Its convolutional feature encoder has seven temporal-convolution blocks and feeds a masked Transformer. Large pretraining scales to roughly 94k hours. The design is important for speaker identification because denoising and overlapped-speech simulation encourage invariance beyond linguistic content.

A downstream speaker system can use weighted sums of WavLM layers, a speaker-specific pooling head, and angular-margin fine-tuning. Frozen-feature extraction is cheap in labels but may leave domain-specific speaker information unused; full fine-tuning improves adaptation but is memory-intensive and risks forgetting.

11.3 Speaker-aware SSL

UniSpeech-SAT explicitly adds utterance-wise contrastive learning and utterance mixing to HuBERT-style pretraining to strengthen speaker information, reporting especially strong speaker-identification-oriented downstream behavior [22]. This illustrates a fundamental point: a universal speech representation may privilege linguistic content unless pretraining includes objectives that preserve identity.

11.4 Foundation-model trade-offs

Foundation encoders reduce label requirements and can improve cross-domain transfer, but they introduce large memory footprints, layer-selection hyperparameters, expensive fine-tuning, and privacy/security concerns if pre-trained on unknown data provenance. For production, compare them against smaller dedicated speaker encoders under equal-domain fine-tuning.

12 Training objectives for discriminative speaker spaces

12.1 Cross-entropy

The baseline speaker-classification objective is

$$\mathcal{L}_{CE} = -\frac{1}{B} \sum_i \log p(y_i | x_i). \quad (22)$$

It is stable and useful for debugging, but does not directly impose a margin in the cosine geometry used at deployment.

12.2 Normalized angular-margin softmax

Let normalized embedding $\bar{z}_i$ and normalized class weight $\bar{w}_j$ satisfy

$$\cos \theta_j = \bar{w}_j^T \bar{z}_i. \quad (23)$$

AAM-Softmax uses the target logit $s \cos(\theta_y + m)$ and non-target logits $s \cos \theta_j$:

$$\mathcal{L}_{AAM} = -\log \frac{e^{s \cos(\theta_y + m)}}{e^{s \cos(\theta_y + m)} + \sum_{j \neq y} e^{s \cos \theta_j}}. \quad (24)$$

The scale s stabilizes softmax optimization; margin m forces angular separation. ECAPA and TitaNet both use angular-margin training in their canonical recipes [16, 17].

12.3 Triplet loss

For anchor a , positive p , and negative n ,

$$\mathcal{L}_{tri} = [d(z_a, z_p) - d(z_a, z_n) + \alpha]_+. \quad (25)$$

Triplet learning aligns training directly with relative distances but depends strongly on negative mining. Deep Speaker is a representative end-to-end triplet system [10].

12.4 GE2E

GE2E forms batches with multiple speakers and multiple utterances per speaker, computes speaker centroids, and encourages each utterance to be close to its own centroid and far from others [15]. It is especially natural for enrollment-style learning because the training unit resembles a small gallery.

12.5 Metric vs classification objectives

Classification objectives scale well to many speakers and often produce excellent embeddings when the final classifier is discarded. Metric objectives align more directly with open-set distance geometry but require carefully structured batches. Hybrid strategies, angular margins, sub-centers, inter-top-k penalties, and large-margin fine-tuning are common in competitive systems and toolkits [27].

13 Speaker-balanced minibatches and augmentation

13.1 Why random utterance sampling is wrong

Speaker datasets are often highly imbalanced in utterances per identity. Uniform utterance sampling over-represents prolific speakers. A better batch samples speakers first, then one or more segments per selected speaker. GE2E-like objectives require multiple utterances per speaker by design.

13.2 Noise and reverberation

The x-vector paper established data augmentation with additive noise and reverberation as a low-cost way to multiply training conditions [11]. Modern recipes use room impulse responses, babble, music/noise mixtures, gain variation, codec simulation, and feature masking.

13.3 Speed perturbation

Speed perturbation modifies duration and pitch. Some recipes treat perturbed versions as pseudo-speaker classes, effectively increasing speaker count. This can help robustness but must be validated because it may encourage sensitivity to speaking rate/pitch if labels are assigned incorrectly.

13.4 Crop length and large-margin fine-tuning

Random 2–4 s crops are common for base training because they enable large batches and teach duration invariance. A late fine-tuning stage can use longer segments, smaller learning rate, larger angular margin, and reduced augmentation. This sharpens class geometry once invariance is learned.

14 Enrollment design

14.1 Multiple utterances reduce template variance

One enrollment clip captures speaker plus session, text, device, and noise. Averaging several normalized embeddings reduces these nuisance effects. VoxBlink2 evaluates 1, 3, and 5 enrollment utterances per speaker and shows

that multiple enrollment improves open-set identification, especially at larger galleries [23].

14.2 Quality-weighted prototypes

A robust prototype can use

$$\mathbf{g} = \frac{\sum_k q_k \bar{\mathbf{z}}_k}{\|\sum_k q_k \bar{\mathbf{z}}_k\|_2}, \quad (26)$$

with q_k based on effective speech duration, SNR, clipping, VAD confidence, or model-predicted uncertainty. Weighting should be learned or validated; ad-hoc quality weights can systematically bias against certain devices or speaking styles.

14.3 Enrollment diversity

Collect different phrases, sessions, and representative devices. Diversity is more valuable than many near-duplicate clips from one recording. Enrollment identity assurance must be stronger than ordinary probe handling because one contaminated template can affect all future decisions.

15 Scoring, back ends, and large-gallery search

15.1 Cosine similarity

Cosine scoring is the default for L2-normalized angular-margin embeddings:

$$s(a, b) = \frac{a^T b}{\|a\|_2 \|b\|_2}. \quad (27)$$

It is simple, fast, and often competitive with learned back ends.

15.2 PLDA

PLDA models within- and between-speaker variability probabilistically and remains useful for classic x-vector systems and domain adaptation. Strong modern margin-trained embeddings may reduce its advantage relative to cosine, but PLDA can still help under specific duration/domain conditions.

15.3 Score normalization

Adaptive S-norm and related cohort methods normalize a trial score relative to impostor cohorts. They can improve verification and sometimes identification calibration, but they add cohort management and compute. They should be evaluated rather than enabled by default.

15.4 Approximate nearest-neighbor retrieval

Exact gallery search costs $O(Nd)$ per probe. For thousands of speakers this is often acceptable; for millions,

an approximate nearest-neighbor (ANN) index such as HNSW or IVF/PQ can retrieve top- K candidates, followed by exact cosine re-ranking and a calibrated unknown threshold. The ANN recall itself must be measured because a retrieval miss cannot be repaired by the downstream threshold.

16 Closed-set evaluation

16.1 Top-1 and Top-k

Top-1 identification accuracy is

$$\text{Acc}_1 = \frac{1}{M} \sum_{p=1}^M \mathbf{1}[\hat{y}_p = y_p]. \quad (28)$$

Top- k counts a hit if the true speaker appears among the k highest-scoring gallery identities. The cumulative match characteristic (CMC) curve plots identification rate versus rank.

16.2 Class-balanced metrics

If probe counts per speaker are unequal, report macro recall, macro F1, and per-speaker error in addition to overall accuracy. A high micro accuracy can hide systematic failures on underrepresented identities.

16.3 Confusion analysis

A confusion matrix is useful for small/moderate rosters. For large rosters, report nearest-impostor pairs and analyze speaker clusters by language, device, duration, and acoustic similarity.

17 Verification metrics as embedding diagnostics

17.1 False acceptance and false rejection

For threshold τ ,

$$P_{FA}(\tau) = P(s \geq \tau \mid \text{impostor}), \quad (29)$$

$$P_{miss}(\tau) = P(s < \tau \mid \text{target}). \quad (30)$$

EER is the point where they are equal. It is threshold-independent in the sense that the threshold is chosen at equality, but it is not an open-set 1: N metric.

17.2 Detection cost

A common detection cost is

$$C_{det}(\tau) = C_{miss}P_{miss}(\tau)P_{tar} + C_{fa}P_{fa}(\tau)(1 - P_{tar}). \quad (31)$$

Minimum DCF optimizes this over τ . VoxSRC and NIST-style evaluations use such operating-cost concepts because practical false-accept and miss costs are asymmetric [6].

17.3 DET and ROC curves

DET plots miss and false alarm in a normal-deviate transform, spreading low-error regions; ROC plots true positive rate versus false positive rate. Both reveal trade-offs hidden by a single EER.

18 Open-set identification metrics

VoxBlink2 explicitly separates gallery G , known probes K , and unknown probes U [23]. For known probe p , let G_p be its correct gallery identity. At threshold θ and rank criterion n ,

$$\text{DIR}(\theta, n) = \frac{|\{p \in K : \text{rank}(p) \leq n, s(p, G_p) \geq \theta\}|}{|K|}. \quad (32)$$

For unknown probes,

$$\text{FAR}(\theta) = \frac{|\{p \in U : \max_{g \in G} s(p, g) \geq \theta\}|}{|U|}. \quad (33)$$

DIR@FAR fixes an operational false-alarm level and reports the corresponding known-speaker identification rate.

18.1 Why gallery size changes the problem

With more enrolled speakers, the maximum impostor score tends to increase because each unknown or known probe competes against more identities. Therefore a threshold calibrated on 60 speakers may fail at 6,000. Every open-set result should report gallery size, enrollment count, number of unknown probes, and target FAR.

19 Datasets

19.1 VoxCeleb impact

VoxCeleb shifted the field from controlled, small speaker corpora toward real-world celebrity speech from video. VoxCeleb2 further increased scale to over one million utterances from over 6,000 speakers and became the default training source for many supervised systems [13].

19.2 VoxSRC as a methodological driver

The VoxCeleb Speaker Recognition Challenge ran annually from 2019 to 2023 with closed/open training, supervised, self-supervised, semi-supervised, verification, and diarization settings. Its retrospective documents the field's transition toward stronger ResNet backbones, data augmentation, score normalization, large-margin training, model fusion, and SSL/domain adaptation [6].

19.3 VoxBlink2 scale and OSSSI

VoxBlink2 extends public speaker scale by roughly two orders of magnitude beyond older identification datasets

Table 1: Representative datasets and their role in deep speaker identification. Counts follow the cited dataset papers/pages and should be interpreted with official split definitions.

Dataset	Speakers	Approx. scale	Role and caution
VoxCeleb1 [12]	1,251	>100k utterances	Landmark in-the-wild speaker ID/SV; verification and identification protocols differ.
VoxCeleb2 [13]	>6,000	>1M utterances	Standard supervised training corpus; large diversity of nationality, accent, age, language, and recording condition.
CN-Celeb [24]	1,000+	multi-genre	Chinese in-the-wild domain and genre stress; useful for domain mismatch.
SITW [25]	many	in-the-wild trials	Verification stress under natural video/audio conditions.
VoxBlink2 [23]	111,284	9,904,382 utterances, 16,672 h	Very-large-scale pretraining and open-set identification; includes S/M/L gallery protocols.
NIST SRE	varies	multi-year	Standard verification operating-cost evaluation; each year has different domains/protocols.
ASVspoof	varies	genuine + spoof	Security benchmark for replay, synthetic/converted, and deepfake attacks; not a normal SID corpus.

and introduces open-set protocols with explicit unknown speakers [23]. It is important because it shifts evaluation from “can two embeddings be verified?” to “can one probe be found among thousands while unknown people are rejected?”

20 Robustness and domain generalization

20.1 Noise and reverberation

Noise changes spectral statistics and can dominate attention; reverberation smears temporal detail. Synthetic augmentation is effective but should be supplemented by real target-domain recordings. Train/dev/test should be stratified by SNR and room/channel where possible.

20.2 Device and codec

Telephony, smartphones, conferencing systems, far-field arrays, and body-worn microphones have different transfer functions. Domain-matched fine-tuning and score calibration often produce larger practical gains than moving to a larger generic encoder.

20.3 Language, accent, and phonetic coverage

Text-independent recognition should not depend on a phrase, but identity cues are observed through phonetic content. Cross-language performance benefits from diverse training and longer utterances. Speaker-aware SSL explicitly attempts to retain identity while learning content-invariant structure [22].

20.4 Emotion, illness, and speaking style

Whispering, shouting, stress, fatigue, and temporary physiological change can shift embeddings. High-stakes deployments should evaluate such strata explicitly and avoid

assuming enrollment from neutral studio speech covers them.

20.5 Short utterances

Short probes contain fewer phonetic events and noisy statistics. Remedies include multi-utterance enrollment, duration-aware calibration, short-crop augmentation, attentive pooling, and minimum speech-duration policies. Report accuracy and DIR separately by duration bins.

21 Open-set few-shot adaptation

Large foundation encoders make it possible to adapt a general representation to a small target roster using only a few examples. Recent SpeakerRPL v2 work explores reciprocal-point learning, logit normalization, adaptive anchors, and model fusion for few-shot open-set identification [29]. Such work is promising but often uses newly defined protocols; reported error reductions should not be compared directly with standard VoxCeleb verification EER.

The fundamental tension is between adapting strongly to a few target speakers and preserving enough global geometry to reject unknown speakers. Overfitting a target classifier can improve closed-set accuracy while worsening open-set tail risk.

22 Security: replay, voice conversion, and deepfakes

Speaker identification is a biometric decision, but speaker discrimination alone is not liveness detection.

22.1 Attack classes

- Replay: play a genuine target recording to the microphone.

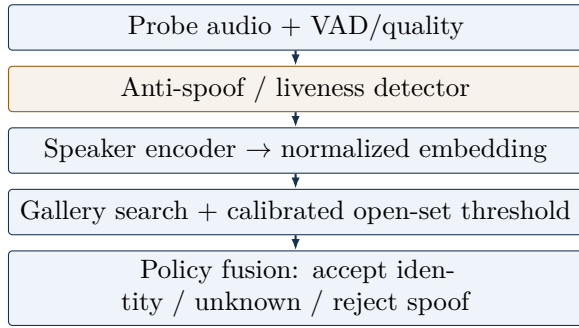


Figure 2: Identity and liveness should be separately trained and separately evaluated.

- Voice conversion: transform an attacker’s voice toward the target.
- Text-to-speech/deepfake: generate target-like speech from text or latent representations.
- Adversarial perturbation: manipulate waveform features to change embedding similarity.

22.2 Tandem identity and spoof decisions

A safe architecture uses an explicit anti-spoof/liveness subsystem before or alongside speaker matching. A high speaker similarity should not override a spoof rejection. ASVspoof evaluations motivate tandem metrics such as t-DCF because countermeasure and ASV errors interact [26].

23 Privacy, fairness, and biometric governance

Speaker embeddings remain biometric templates even when raw audio is deleted. They can link recordings and may encode demographic, language, health, or emotional attributes. Systems should therefore use encryption, role-based access, retention/deletion policies, template versioning, audit logs, and explicit consent/lawful basis where required.

Fairness evaluation should compare false accept, false reject, and identification rates across relevant subgroups, devices, languages, durations, and environments. A single global threshold can create unequal error rates if score distributions differ by subgroup. Threshold adjustments themselves raise policy questions and should be documented.

Model updates create a template-compatibility problem: embeddings from encoder version v_1 may not be comparable to v_2 . Production galleries need re-embedding or explicit backward-compatibility validation.

24 Production architecture and toolkits

24.1 Service decomposition

A maintainable speaker-identification service separates waveform ingestion, resampling/VAD, quality checks, anti-spoofing, embedding extraction, template storage, vector search, calibration/policy, and monitoring. Separation permits independent model upgrades and simplifies security review.

24.2 WeSpeaker and 3D-Speaker

WeSpeaker provides speaker embedding models, losses, scoring back ends, supervised and SSL recipes, and CPU/GPU deployment paths including ONNX/TensorRT-oriented runtime support [27]. 3D-Speaker extends the ecosystem toward supervised/SSL acoustic models and multimodal speaker verification/diarization [28]. These toolkits are useful reference implementations but should not substitute for project-specific reproducibility records.

24.3 Latency and throughput

End-to-end latency can be decomposed as

$$T_{total} = T_{audio} + T_{VAD} + T_{feat} + T_{enc} + T_{search} + T_{policy}. \quad (34)$$

For live systems, audio acquisition often dominates; for archived clips, encoder and vector search dominate. Benchmark batch-one latency, not only large-batch GPU throughput.

25 Architecture comparison

26 What the > 500-paper corpus says about field direction

The 2026 corpus-level analysis reports a clear publication transition from x-vector research to ResNet/ECAPA systems and then to Transformer and hybrid systems, while ResNet/ECAPA remains the most extensively studied family [1]. This suggests five durable trends.

26.1 Trend 1: multi-scale feature learning

Speaker traits exist at different time scales: glottal cycles, phoneme-scale articulation, syllabic rhythm, and utterance-level prosody. Res2Net, ECAPA, ERes2Net, and multi-layer aggregation directly reflect this requirement.

26.2 Trend 2: global context without abandoning local inductive bias

Pure attention is powerful but expensive; pure convolution is efficient but indirect for long-range interaction.

Table 2: Qualitative architecture comparison for speaker identification. “Compute” is relative and depends strongly on implementation and input duration.

Family	Local modeling	Global context	Compute	Best use
x-vector TDNN	strong temporal	pooling only	low	Mature baseline, CPU, legacy interoperability.
ECAPA-TDNN	multi-scale temporal + channel attention	SE + attentive pooling	medium	Default supervised speaker embedding baseline.
ResNet/ERes2Net	strong 2D time-frequency hierarchy	pooling/fusion	medium-high	High-accuracy server systems; large-data scaling.
TitaNet	depth-wise 1D temporal	SE + channel attention pooling	low-medium	Efficient deployment and diarization/SID throughput.
CAM++	deep dense TDNN + front-end CNN	context-aware masks + multi-granularity pool	low-medium	Accuracy/efficiency balance under constrained compute.
MFA-Conformer	convolution + attention	explicit self-attention	high	Large server models, long context, hybrid representation.
WavLM / SSL	learned waveform + Transformer	explicit global attention	high/very high	Low-label transfer, foundation-style multi-task systems.

Conformer and hybrid CNN-Transformer systems combine both. Hybrid growth in the recent corpus reflects this compromise.

26.3 Trend 3: data scale and model scale co-evolve

VoxCeleb enabled deep CNN speaker embeddings; VoxBlink2 demonstrates another order-of-magnitude scale increase. Larger models become more useful when speaker diversity and session diversity increase. Without data diversity, scale can amplify shortcut learning.

26.4 Trend 4: supervision is broadening

The field is moving from fully supervised speaker classification toward self-supervised, semi-supervised, speaker-aware pretraining, and few-shot open-set adaptation [5, 29]. The target is not merely lower EER; it is reusable identity geometry under label scarcity and domain shift.

26.5 Trend 5: evaluation is moving toward application realism

Verification remains the dominant benchmark, but open-set identification, large galleries, unknown speakers, spoof attacks, and multimodal evidence are increasingly explicit. VoxBlink2 is a notable step because it makes gallery size and unknown rejection part of the standard task definition.

27 Reported performance snapshots: use with caution

Numbers should be read as demonstrations of what a complete recipe can achieve, not as isolated architecture rankings. The fair question is: under equal data, equal augmentation, equal scoring, and similar parameter bud-

get, which design performs better on the target domain and operating point?

28 A recommended reference system

28.1 Baseline configuration

For a new text-independent open-set speaker-identification project:

- 16 kHz mono input; deterministic resampling.
- VAD plus minimum speech duration and clipping/SNR checks.
- 80-bin log-Mel FBANK, 25 ms window, 10 ms hop.
- ECAPA-TDNN or strong ResNet-family encoder.
- 192–256 dimensional L2-normalized embedding.
- AAM-Softmax with speaker-balanced batches.
- Noise, babble, RIR/reverb, gain, codec, and validated speed perturbation.
- Multi-utterance enrollment (3–5 diverse clips when possible).
- Cosine scoring; ANN retrieval only if gallery scale requires it.
- Unknown threshold calibrated on held-out unknown speakers at target FAR.
- Separate anti-spoof/liveness gate.

28.2 Training stages

Stage 1: train the speaker encoder with 2–4 s random crops and broad augmentation.

Stage 2: select the best development checkpoint.

Stage 3: optional large-margin fine-tuning with longer crops, smaller learning rate, larger margin, and reduced augmentation.

Table 3: Selected paper-reported results. They are illustrative and not a fair leaderboard because training data, model size, post-processing, and protocol differ.

System	Metric/protocol	Reported value	Interpretation
ECAPA-TDNN large [16]	VoxCeleb1 EER	0.87%	Strong 2020 speaker-specialized reference with adaptive score normalization.
TitaNet [17]	VoxCeleb1 trial EER	0.68%	Efficient depth-wise separable model; paper-specific multi-corpus recipe.
MFA-Conformer [20]	VoxCeleb1-O EER	0.64%	Demonstrates local/global hybrid modeling.
CAM++ [18]	VoxCeleb-O EER	0.73%	Efficiency-focused D-TDNN/context-aware masking recipe.
VoxBlink2 large system [23]	VoxCeleb1-O EER	0.170%	Very large speaker pretraining and model scale; not architecture-only evidence.

Stage 4: enroll development speakers, calibrate thresholds, and freeze evaluation settings before test.

28.3 Evaluation matrix

At minimum report:

- closed-set Top-1/Top-5 and macro recall;
- verification EER/minDCF as embedding diagnostics;
- open-set DIR@FAR at several FAR levels;
- gallery sizes representative of deployment;
- 1, 3, 5 enrollment utterances;
- probe duration bins;
- SNR/device/language/domain strata;
- anti-spoof tandem error or attack-specific rates;
- model size, peak memory, batch-one latency, and RTF.

29 Experimental design and reproducibility

29.1 Avoid identity and session leakage

Training and open-set test speakers must be disjoint. If clips are segmented from the same original recording, do not distribute sibling segments across train/dev/test. Session leakage can make a model memorize channel/background rather than identity.

29.2 Ablation sequence

A disciplined sequence is:

1. x-vector or ECAPA baseline with CE;
2. switch to AAM-Softmax;
3. add augmentation;
4. compare pooling;
5. add multiple-enrollment prototypes;
6. calibrate open-set threshold;

7. large-margin fine-tuning;
8. compare efficient encoder;
9. compare SSL/foundation encoder;
10. compare ANN search and quality weighting only after embedding accuracy is stable.

Changing one major factor at a time avoids attributing data or score-normalization gains to the backbone.

29.3 Uncertainty

Report confidence intervals, preferably by bootstrapping speakers or trials rather than individual frames. Small EER differences can be statistically meaningless when trials are correlated by speaker/session.

30 Failure analysis

30.1 Unknown false accepts

For false accepts, inspect: top impostor identity, duration, SNR, language/accent, device, same-background/channel cues, spoof status, and whether the gallery has acoustically similar speakers. Extreme-tail impostor scores matter more than average inter-speaker separation.

30.2 Known misidentification

Compare true-speaker score s_{true} with top impostor score s_{imp} . If both are low, suspect low quality or domain mismatch. If both are high, suspect confusable speakers or insufficiently diverse enrollment. If the true speaker ranks highly but not first, metric learning or gallery prototypes may help more than VAD.

30.3 Enrollment contamination

Enrollment errors are particularly dangerous. A contaminated prototype can cause persistent future false accepts/rejects. Use stronger identity assurance and quality checks during enrollment than during normal probe inference.

31 Open research questions

31.1 Large-gallery tail calibration

As galleries move from thousands to millions of identities, the maximum impostor distribution and approximate-search error become central. Research needs conditional thresholds, extreme-value calibration, efficient hard-impostor mining, and confidence estimates that remain stable under gallery growth.

31.2 Few-shot and continual enrollment

Real systems add/remove speakers without full retraining. Future methods should adapt prototypes or lightweight adapters while preserving unknown rejection and old-speaker geometry.

31.3 Speaker-aware foundation learning

General speech foundation models must balance content, identity, emotion, and environment. Speaker-aware pre-training objectives, denoising, multi-view contrast, and selective layer fusion remain active research areas [5, 21, 22].

31.4 Deepfake-robust identity

Generative speech quality is improving rapidly. Identity systems need representations that separate real target speech from high-similarity synthetic/replayed speech without unfairly penalizing atypical genuine speech. Joint training must avoid anti-spoof features corrupting identity geometry.

31.5 Privacy-preserving templates

Revocable transformations, secure matching, federated learning, and leakage auditing are needed because speaker embeddings are linkable biometric data. A compromised speaker embedding cannot be “changed” like a password without a transform or new model version.

31.6 Multimodal identity

Audio-visual and semantic cues can improve robustness in meetings/video, as reflected by large audio-visual corpora and multimodal toolkits [23, 28]. The challenge is graceful degradation when one modality is missing or spoofed.

multi-scale modeling, global context, label efficiency, and deployment cost.

For speaker identification specifically, the encoder is only one component of a successful system. A high-quality solution must pair the representation with disciplined data splitting, speaker-balanced training, attentive aggregation, margin-based geometry, diverse enrollment, similarity search, open-set threshold calibration, gallery-size-aware evaluation, and explicit anti-spoof controls. Closed-set Top-1, verification EER, and open-set DIR@FAR answer different questions and should never be substituted for one another.

The most defensible practical baseline in 2026 remains a speaker-specialized supervised embedding model such as ECAPA-TDNN or a strong ResNet-family encoder, trained with AAM-Softmax and multi-condition augmentation, then deployed with normalized enrollment prototypes and cosine scoring. Efficient alternatives such as CAM++ and TitaNet deserve early benchmarking under hardware constraints; Conformer/SSL models deserve early benchmarking where large-scale pretraining or label scarcity justifies their complexity. Open-set identification should be treated as the target task whenever unknown speakers can occur, and its acceptance threshold must be calibrated to a stated FAR under a realistic gallery size.

32 Conclusions

The > 500-document evidence base shows a field that has converged on deep speaker embeddings as the central abstraction. The progression from x-vectors to ECAPA/ResNet, efficient depth-wise networks, Conformer hybrids, and SSL/foundation models is best understood as a sequence of attempts to improve invariance,

A Compact mathematical reference

A.1 Speaker embedding

$$\bar{z} = f_{\theta}(x) / \|f_{\theta}(x)\|_2.$$

A.2 Cosine score

$$s(a, b) = a^T b / (\|a\|_2 \|b\|_2).$$

A.3 Prototype

$$g_i = \text{norm} \left(\sum_k q_{ik} \bar{z}_{ik} \right).$$

A.4 AAM-Softmax

$$\mathcal{L} = -\log \frac{e^{s \cos(\theta_y + m)}}{e^{s \cos(\theta_y + m)} + \sum_{j \neq y} e^{s \cos \theta_j}}.$$

A.5 Open-set decision

$$i^* = \arg \max_i s_i, \quad \hat{y} = \begin{cases} i^*, & s_{i^*} \geq \tau \\ \text{unknown}, & s_{i^*} < \tau. \end{cases}$$

A.6 DIR and FAR

$$\text{DIR}(\theta, 1) = \frac{|\{p \in K : \text{rank}(p) = 1, s(p, G_p) \geq \theta\}|}{|K|},$$

$$\text{FAR}(\theta) = \frac{|\{p \in U : \max_g s(p, g) \geq \theta\}|}{|U|}.$$

B Reproducibility checklist

- Exact train/dev/test speaker IDs and session-leakage policy.
- Sample rate, channel conversion, resampler, VAD, minimum duration.
- Feature dimensions, window/hop, FFT, normalization statistics.
- Full augmentation sources, probabilities, SNR ranges, RIR/codec/speed settings.
- Encoder topology, channels/depth, pooling, embedding dimension, parameter count.
- Loss formula, margin/scale, speaker sampling, batch composition.
- Optimizer, LR schedule, weight decay, gradient clipping, mixed precision.
- Crop duration and large-margin fine-tuning settings.
- Enrollment count, prototype rule, quality filtering.

- Scoring back end, score normalization/cohorts, ANN index parameters.
- Unknown development set, target FAR, threshold-selection algorithm.
- Gallery size, known/unknown probe counts, duration/SNR/device strata.
- Anti-spoof model and tandem policy.
- Hardware, precision, batch-one latency, RTF, peak memory.
- Encoder/template version and backward-compatibility policy.

C Evidence limitations

This paper is a synthesis rather than a new controlled benchmark. The breadth claim refers to the 929-paper full-text review corpus reported by Khamsehashari et al. [1]; this synthesis did not independently download and re-read every one of those 929 PDFs. Direct primary-paper analysis was concentrated on representative architecture, dataset, objective, and open-set benchmark papers. Reported results are preserved as paper-specific evidence and are not normalized into a synthetic leaderboard.

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